

Narrow-band Weil audit

What the frequently-selected mincut would prove

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Abstract

The frequently-selected Lean route does not currently prove RH. Its formal endpoint is nonnegativity of a three-channel object `weilForm` on dyadic piecewise-linear autocorrelations. The spectral/zero side and the density/continuity passage to the standard Weil criterion are explicitly absent; the path also consumes one typed classical `sorryAx` and one named, unproved Hankel/read bridge. After those gaps are closed, the correct classical translation is not a harmless narrow-band statement: it is positivity of the compact-window Weil form on an unbounded cofinal sequence of support windows, hence—by nestedness—the full Weil criterion.

Numerically, the `r459` selected builder has effective autocorrelation support

$$U_k = N_k \Delta_k = \frac{k \log 2}{2}.$$

Every measured member $k = 5, \dots, 12$ lies beyond both the classical prime-blind range $U < \log 2$ and the recent `arXiv:2608.24827` certificate $U \leq 1.6$. The `r459` race 0.61–0.76 therefore measures a genuinely conditional tail, subject to the still-open dictionary. Verdict: `FORK_SHARPENED`. No RH claim.

Status. Consistency and solution probe only. No new theorem about ζ . No L^* claim and no $R^\dagger$ claim. No RH claim and no anti-RH claim.

1 Q1: the statement actually present in Lean

1.1 The test class

Proposition 1 (Exact Lean class). *A `GridElement` consists of*

$$(n_g, m_f, x), \quad x : \{0, \dots, n_g - 1\} \longrightarrow \mathbb{R}, \quad D_0 = 2^{-m_f}.$$

It defines the autocorrelation coefficients

$$a_d = D_0 \sum_i x_i x_{i+d}$$

and the even piecewise-linear interpolant g_x through a_d on the dyadic grid. Its support is contained in $[-n_g D_0, n_g D_0]$.

This is literal from `Elementwise.lean:248–253` (fields), `260–273` (D_0 and autocorrelation), `293–301` (even piecewise-linear interpolant), and `303–334` (support and evenness). There is no smoothness field. The induced g_x is continuous and piecewise linear, generally not C^∞ .

Classically $g_x = h_x * \hat{h}_x$, where h_x is the corresponding dyadic step function. Therefore

$$\hat{g}_x(t) = |\hat{h}_x(t)|^2 \geq 0.$$

Thus the relevant positivity is Fourier positivity of the *autocorrelation*; it is not an additional assumption on an arbitrary test function.

1.2 The three-channel value

The arithmetic channel is genuinely transcribed:

$$\text{combRead}(a, g_x) = \sum_{n \in \text{windowAtoms}(a)} \frac{2\Lambda(n)}{\sqrt{n}} g_x(\log n),$$

$$\text{fullRead}(a, m, g_x) = \text{archRead} - \text{combRead} + \text{poleRead}.$$

See `Elementwise.lean:361–391` and `711–728`. The exact finite stabilization of the comb is proved at `Elementwise.lean:396–430`. The archimedean read and its Weil pairing remain opaque (`Elementwise.lean:553–565`), and their identification is the typed `sorry arch_elementwise_stabilization` (`Elementwise.lean:676–698`).

1.3 Selected covering and endpoint

The sequence is exactly

$$a_k = 2^k, \quad r_k = \lfloor \sqrt{k} \rfloor, \quad m_k = k2^{r_k} - 1, \quad \Delta_k = 2^{-r_k} \log 2,$$

at `Selected.lean:293–343`. Both a_k and m_k tend to infinity and `selected_covers` supplies, for each fixed grid element, one sufficiently large selected index (`Selected.lean:347–418`). There is no fixed support ceiling in this quantifier: the `steps` field is unbounded and the sequence is cofinal.

The extraction `weil_nonneg_of_frequently_plain` (`FrequentlySelected.lean:345–355`) takes one index that is both covering and good. The named mincut is

$$\exists^\infty k : \quad R^\dagger(W_k^{\mathbb{R}}) \succeq \tfrac{1}{2}I$$

(`FrequentlySelected.lean:260–285`). The real-window Loewner equivalence to $A_{\text{cap}} \succeq 0$ is proved (`FrequentlySelected.lean:370–380`). The remaining implication

$$A_{\text{cap}} \succeq 0 \implies \text{fullRead}(a_k, m_k, g_x) \geq 0$$

is only the named `Prop`

$$\text{SelectedACapPsdImpliesPlainReads}$$

at `FrequentlySelected.lean:382–409`.

Proposition 2 (Formal endpoint). *Even after assuming the mincut and the named bridge, the theorem historically called `rh_of_frequently_selected` concludes only*

$$\forall f : \text{GridElement}, \quad 0 \leq \text{weilForm}(f).$$

It does not conclude the Riemann hypothesis.

See `FrequentlySelected.lean:430–449`. The source explicitly says that the spectral/zero side is not formalized (`Elementwise.lean:61–77`, `721–726`), and that the density/continuity step off the native class is deliberately not formalized (`Elementwise.lean:88–93`). The audit records `sorryAx` on this path and names the read-identification remainder (`Audit.lean:280–293`, `323–339`). Consequently the premise “Lean mincut $\Rightarrow$ RH, sorry-free” is false in the current tree.

1.4 A second dictionary warning

The operational `r459` builder sets $M = m_k + 1$ and $N_k = M/2$. The Lean object `ExactFold`, however, first maps exact $\log n$ nodes by a cosine and merges equal images (`Selected.lean:160–179`; `Source.lean:400–424`); its `cap` is $(S + 1)/2$ for that exact folded support (`Selected.lean:242–246`). The equality of this exact-fold cap with the `r459` $M/2$ tent/DFT cap is not a theorem. This is part of the same open Hankel/read dictionary and must be closed before builder numerics can be promoted to the Lean mincut.

2 Q2: the unconditionedness fork

2.1 What is unconditionally known

Use the additive-log form of Weil’s criterion. For a compactly supported base function h , let $g = h * \tilde{h}$. Weil positivity is

$$Q_W(h) := W(g) \geq 0.$$

For $h \in C_c^\infty(\mathbb{R})$, positivity for every h is equivalent to RH (Weil 1952; modern formulations use this nonnegative-definiteness of the Weil distribution).

Bombieri’s Theorem 12 proves strict positivity when the support of h is contained in an interval of length

$$U < \log 2.$$

After translation take $\text{supp } h \subset [-U/2, U/2]$; then $\text{supp}(h * \tilde{h}) \subset [-U, U]$, so the von-Mangoldt sum vanishes. This is the precise Yoshida/Bombieri small-support result, not a Fourier-support statement with the number 2.

A recent preprint, arXiv:2608.24827 (August 25, 2026), claims the certified bound

$$Q_W(h) \geq 8.9 \cdot 10^{-18} \|h\|_2^2, \quad \text{supp } h \subset [-0.8, 0.8],$$

hence autocorrelation support $U \leq 1.6$. This extends the classical support length $\log 2$ but is a recent computational preprint, not the classical Yoshida/Bombieri theorem.

Remark 1 (The alleged $(-2, 2)$ theorem is the wrong regime). *There is no standard unconditional theorem saying that “ $\text{supp } f \subset (-2, 2)$ implies Weil positivity for ζ ” in the normalization above. Support restrictions such as $(-2, 2)$ occur in scaled one-/two-level density and pair-correlation formulae; their coordinate is normalized by a conductor or height. They are not a universal compact-log-support certificate for the Weil quadratic form. Fejér kernels supply nonnegative Fourier transforms, but do not make the prime term of the Weil formula nonnegative for arbitrary support.*

2.2 Numerical location of the selected windows

In the r459 reference builder,

$$N_k = \frac{m_k + 1}{2} = k2^{r_k - 1}, \quad U_k = N_k \Delta_k = \frac{k \log 2}{2}.$$

U_k is the support half-width of the autocorrelation, equivalently the length of the underlying step-function interval. In the centered convention $\text{supp } h \subset [-L_k, L_k]$,

$$L_k = \frac{U_k}{2} = \frac{k \log 2}{4}.$$

k	m_k	N_k	Δ_k	U_k	L_k	J_P	$n_{1.6}$	tail	race
5	19	10	.1732868	1.732868	.866434	3	9	1	.675060
6	23	12	.1732868	2.079442	1.039721	3	9	3	.694907
7	27	14	.1732868	2.426015	1.213008	3	9	5	.628736
8	31	16	.1732868	2.772589	1.386294	3	9	7	.606566
9	71	36	.0866434	3.119162	1.559581	7	18	18	.635369
10	79	40	.0866434	3.465736	1.732868	7	18	22	.757783
11	87	44	.0866434	3.812309	1.906155	7	18	26	.715782
12	95	48	.0866434	4.158883	2.079442	7	18	30	.693564

Here $J_P = 2^{r_k} - 1$ is the r459 prime-blind prefix marker. $n_{1.6} = \lfloor 1.6/\Delta_k \rfloor$ is the last depth covered by the recent $U \leq 1.6$ certificate, and “tail” is $N_k - n_{1.6}$. Every measured member is outside the recent range: the crossing occurs between $k = 4$ ($U_4 = 1.38629$) and $k = 5$ ($U_5 = 1.73287$). The classical crossing occurs already between $k = 2$ ($U_2 = \log 2$ at the endpoint) and $k = 3$.

Proposition 3 (Fork verdict). *Alternative (a) does not occur on the measured ladder. Alternative (b) starts at its first member, $k = 5$. The r459 race is therefore a finite measurement in the conditional support tail, not a consequence of the classical narrow-band theorem.*

2.3 Literal classical translation of the mincut

Literal translation. For infinitely many unbounded selected indices k , the augmented dual-resolvent cone of the full selected prime window is semidefinite at the half-filling cap. If the missing Hankel/read identification is valid, this says that the Riemann–Weil quadratic functional is nonnegative on every dyadic step-function autocorrelation resolved and covered by that selected window. Since the selected supports are cofinal, every compactly supported dyadic autocorrelation is covered by one good window. After the missing continuity/density and spectral explicit-formula identifications, this is *cofinal compact-window Weil positivity*.

Define the classical compact-window floor

$$\lambda_*(L) = \inf_{\substack{0 \neq h \in L^2(\mathbb{R}) \\ \text{supp } h \subset [-L, L]}} \frac{Q_W(h)}{\|h\|_2^2}.$$

The honest classical name for the completed mincut is

$$\lambda_*(L_k) \geq 0 \quad \text{for an unbounded sequence } L_k.$$

Because the spaces are nested, $\lambda_*(L)$ is nonincreasing. Hence positivity on any unbounded cofinal subsequence implies positivity on every finite L . This is not a weaker “positive-density” fragment after the dictionary is closed; it is the full Weil criterion.

3 Q3: best complete route

Step	Task	Honest status
1	Prove that the exact folded Lean window and the tent/DFT r459 window have the same relevant Hankel form and cap; repair any dimension or mesh-normalization mismatch.	Open, local and decisive. Finite algebra after a faithful fold transcription, but not present.
2	Prove <code>SelectedACapPsdImpliesPlainReads</code> , including bordered-to-plain compression and the exact arch/border read.	Open, plausibly mechanizable; this is the current load-bearing bridge, not the $R^\dagger$ Loewner algebra.
3	Replace <code>arch_elementwise_stabilization</code> ’s sorry by the classical digamma/Mellin identity.	Known mathematics, formalization work.
4	Formalize dyadic step density, continuity of the localized Weil form, and the Guinand–Weil spectral/zero identity.	Known mathematics, substantial Lean work. Only after this step may the endpoint be called an RH criterion.
5	Certify $\lambda_*(L) \geq 0$ at additional fixed L by interval matrices. Reproduce the claimed $L = .8$ certificate, then target $L_5 = .866434$.	Finite and in principle feasible; margins may be extremely small. Each fixed L is only a finite RH fragment.
6	Prove nonnegativity on unbounded L_k (equivalently, the frequently-selected mincut after Steps 1–4).	Open-hard: equivalent to RH. No known density or subsequence theorem supplies it.

Why zero-density theorems do not close Step 6. Results that put a positive proportion of zeros on the critical line control averages or restricted zero sets. Weil positivity requires nonnegativity against every compactly supported autocorrelation. A single off-line zero can create a negative direction once the support space is large enough. Thus “many zeros on the line” does not imply the needed cofinal window positivity.

Why Li/Bombieri–Lagarias do not close Step 6. Li’s theorem says RH is equivalent to $\lambda_n^{\text{Li}} > 0$ for every $n \geq 1$. Bombieri–Lagarias derive these coefficients from the Guinand–Weil functional and generalize the zero-set criterion. This is an exact coordinate bridge, useful for cross-checking normalizations and constructing test vectors. It does not make the positivity unconditional; proving all Li coefficients positive is again RH.

Useful diagnostic result. Bombieri’s variational analysis shows that, under a finite off-line-zero hypothesis, sufficiently rich truncations acquire negative eigenvalues whose count detects the failures. This is valuable for red-teaming the selected matrices, but it is not a positive lower bound on a cofinal family.

4 Conclusion

The chain is consistent because it is not presently a closed RH proof and because its measured windows are not in the unconditional narrow-band regime. The strongest honest externalization is:

The selected $R^\dagger$ mincut is a proposed finite-matrix realization of cofinal compact-window Weil positivity. Its Loewner algebra is formalized, but the exact Hankel/read dictionary, one classical archimedean identity, and the density/spectral completion remain open. On the r459 builder the tested windows begin just beyond the current $U = 1.6$ compact-support certificate and stay positive with race 0.61–0.76 through $k = 12$.

Verdict: FORK_SHARPENED. No RH claim. No anti-RH claim.

References

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